Basic principle of generation of Alternating and Direct Current, Introduction to phasor algebra as applied to A.C. circuit analysis. Solution of A.C. circuits: Series, Parallel and Series-Parallel circuit, R.L.C circuits series and parallel resonance. Applications of Networks theorems to A.C. circuits.

The magnetic intensity, flux/density, magnetic effects of Electric current, Magnetic circuit concepts, BH curves, characteristics of magnetic materials, magnetic force and its utilization, Hystersis and eddy current losses, magnetic circuit with A.C. and D.C. excitation.

Recommended Texts:

1. C. K. Alexander and M. N. O. Sadiku, *Fundamentals of Electric Circuits*, 6th ed. McGraw Hill, 2016
2. R. L. Boylestad, *Introductory Circuit Analysis*, 14th ed. Pearson Education India, 2022

CSE 4107 Structured Programming I Credit: 3.00

Introduction, Programming Concepts, Algorithm and Logic, Constants, Variables, Keywords and Data Types, Operators and expressions, Managing Input and Output Operations, Decision Making and Branching, Decision Making and Looping, Arrays, Multi-dimensional Arrays, Strings, User defined functions, Recursion, Structures and Unions, File Management in C, Pointers, Dynamic Memory Allocation and Linked List, The Preprocessor and some advanced topics, Advanced data types and operators.

Recommended Texts:

1. H. Schildt, *Teach Yourself C*, 3rd ed. McGraw-Hill, 1997
2. E. Balagurusamy, *Programming In Ansi C*, 8th ed. Noida, Uttar Pradesh, India: McGraw Hill Education, 2019

SWE 4101 Introduction to Software Engineering Credit: 3.00

Basic Computer Concepts, Concepts in Hardware and Software.

Basic Computer Organization: Processor and Memory, Secondary Storage Devices, Input-Output Devices, Networking, Introduction to Web and other emerging technologies such as Blogs, Wiki, RSS, Podcasting, Cloud applications.

Computer Software: Programming Languages, Compiler, Assembler, Linker.

Software Engineering: Software Development Life Cycle, Introduction to Software Process Models, Software Requirements Analysis, Software Documentation, Introduction to Software Design, Testing, Deliverables and Maintenance.